Esau Hervert
Math 6313

Homework 6.1

Problem 18

Prove or disprove: If n is a divisor of m, then each zero of T_n is a zero of T_m.

$$m = n * k$$

$$T_2(x) = 2x^2 - 1, T_4(x) = 8x^4 - 8x^2 + 1$$

$$\psi = x^2$$

$$T_2(\psi) = 2\psi - 1 \Rightarrow \psi = \frac{1}{2}$$

$$\therefore x_2^1 = \frac{\sqrt{2}}{2}, x_2^2 = -\frac{\sqrt{2}}{2}$$

$$T_4(\psi) = 8\psi^2 - 8\psi + 1 \Rightarrow \psi = \frac{8 \pm \sqrt{64 - 4(8)(1)}}{2 * 8} = \frac{8 \pm \sqrt{32}}{16} = \frac{1}{2} \pm \frac{\sqrt{2}}{4}$$

$$\therefore x_4^1 = \frac{1}{2}\sqrt{2 + \sqrt{2}}, x_4^2 = -\frac{1}{2}\sqrt{2 + \sqrt{2}}, x_4^3 = \frac{1}{2}\sqrt{2 - \sqrt{2}}, x_4^4 = -\frac{1}{2}\sqrt{2 - \sqrt{2}}$$

Note that the roots of T_2 are not the roots of T_4 but $2|4$, thus this is false.

Problem 24

The formula for the leading coefficients in T_n is 2^{n-1}. What is the formula for the coefficients of x^{n-2}? What about x^{n-1}?

$$T_{n+1} = 2xT_n - T_{n-1}$$

$$T_0 = 1$$

$$T_1 = x$$

$$T_2 = 2x^2 + (0)x - 1$$

$$T_3 = 4x^3 + (0)x^2 - 3x$$

$$T_4 = 8x^3 + (0)x^2 - 8x + 1$$

$$T_5 = 16x^4 + (0)x^3 - 20x^2 + 5x$$

$$\ldots$$

$$T_n = 2^{n-1}x^n + a_n x^{n-1} + b_n x^{n-2} + p_{n-3}(x)$$

Note:

$$a_n = 0$$

$$\Rightarrow T_n = 2^{n-1}x^n + (0)x^{n+1} + b_n x^{n-2} + p_{n-3}(x)$$

Using our relation between T_n and T_{n-1}, T_{n-2}:

$$T_{n+1} = 2x(2^{n-1}x^n + b_n x^{n-2} + p_{n-3}(x)) - (2^{n-2}x^{n-1} + b_{n-1}x^{n-3} + p_{n-4}(x))$$

$$\Rightarrow T_{n+1} = 2^n x^{n+1} + 2b_n x^{n-1} + p_{n-3}(x) - 2^{n-2}x^{n-1} - b_{n-1}x^{n-3} + p_{n-4}(x)$$

$$\Rightarrow T_{n+1} = 2^n x^{n+1} + x^{n-1}(2b_n - 2^{n-2}) + P_{n-2}(x)$$

$$\Rightarrow b_{n+1} = 2b_n - 2^{n-2}$$

$$b_3 = 2(-1) - 2^{2-2} = -3$$

$$b_4 = 2(2(-1) - 2^{2-2}) - 2^{3-2} = 4(-1) - 2^{3-2} - 2^{3-2} = -2^2 - 2^{3-2}(2) = -4 - 4 = -8$$

$$b_5 = 2(-2^2 - 2^{4-2}) - 2^{4-2} = -2^3 - 2^{5-2} - 2^{4-2} = -2^3 - 2^{4-2}(2+1) = -2^3 - 2^{4-2}(3) = -16$$

$$\therefore b_n = -2^{n-2} - 2^{n-3}(n-2), n \geq 2$$

Claim:

$$T_n = 2^{n-1}x^n + (0)x^{n-1} + \left(-2^{n-2} - 2^{n-3}(n-2)\right)x^{n-2} + p_{n-3}(x)$$

Note that $p(x)$ is 0 when n is less than 3.

$$T_2 = 2x^2 + 0x + \left(-2^0 - 2^{-1}(2-2)\right) = 2x^2 - 1$$

Thus, the base case holds. Now for the induction step:

$$T_{n+1} = 2xT_n - T_{n-1}$$

$$\Rightarrow T_{n+1} = 2x\left(2^{n-1}x^n + (0)x^{n-1} + \left(-2^{n-2} - 2^{n-3}(n-2)\right)x^{n-2} + p_{n-3}(x)\right) - \left(2^{n-2}x^{n-1} + (0)x^{n-2} + \left(-2^{n-3} - 2^{n-4}(n-3)\right)x^{n-3} + p_{n-4}(x)\right)$$

$$\Rightarrow T_{n+1} = \left(2^n x^{n+1} + (0)x^n + \left(-2^{n-1} - 2^{n-2}(n-2)\right)x^{n-1} + p_{n-2}(x)\right) - \left(2^{n-2}x^{n-1} + (0)x^{n-2} + \left(-2^{n-3} + 2^{n-4}(n-3)\right)x^{n-3} + p_{n-4}(x)\right)$$

$$\Rightarrow T_{n+1} = x^{n+1}2^n + (0)x^n + x^{x-1}(-2^{n-1} - 2^{n-2}(n-2) - 2^{n-2}) - \tilde{p}_{n-2}(x)$$

$$\Rightarrow T_{n+1} = x^{n+1}2^n + (0)x^n + x^{x-1}\left(-2^{n-1} - 2^{n-2}(n-1)\right) - \tilde{p}_{n-2}(x)$$

Thus:

$$a_{n+1} = 0$$

$$b_{n+1} = -2^{n-1} + 2^{n-2}(n-1)$$

Problem 28

Let p_k be the polynomial of degree $\leq k$ such that $p_k(x_i) = y_i$ for $0 \leq i \leq k$. Prove that $p_k = p_{k-1}$ if and only if $p_{k-1}(x_k) = y_k$

($\Rightarrow$)

$$p_k = p_{k-1} \Rightarrow y_k = p_k(x_k) = p_{k-1}(x_k)$$

$$\therefore p_{k-1}(x_k) = y_k$$

($\Leftarrow$)

$$p_{k-1}(x_k) = y_k$$

$$p_k(x) = p_{k-1}(x) + F[x_0, x_1, \dots, x_k] * \prod_{n=0}^{k-1}(x - x_n)$$

$$p_k(x_j) = p_{k-1}(x_j), j = 0,1, \dots, k-1$$

$$p_{k-1}(x_k) = y_k$$

$$\Rightarrow y_k = p_k(x_k) = p_{k-1}(x_k) + F[x_0, x_1, \dots, x_k] * \prod_{n=0}^{k-1}(x_k - x_n) = y_k + F[x_0, x_1, \dots, x_k] * \prod_{n=0}^{k-1}(x_k - x_n)$$

$$\Rightarrow y_k = y_k + F[x_0, x_1, \dots, x_k] * \prod_{n=0}^{k-1}(x_k - x_n) \Rightarrow F[x_0, x_1, \dots, x_k] = 0$$

This is true since x_k is not a root of the polynomial $\prod_{n=0}^{k-1}(x_k - x_n)$, thus:

$$p_k(x_k) = p_{k-1}(x_k)$$

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Problem 7

Determine all values of a, b, c, d, e for which the following function is a cubic spline:

$$S(x) = f(x) = \begin{cases} a(x-2)^2 + b(x-1)^3 & x \in (-\infty, 1] \\ c(x-2)^2 & x \in [1,3] \\ d(x-2)^2 + e(x-3)^3 & x \in [3, \infty) \end{cases}$$

$$s_1(x) = a(x-2)^2 + b(x-1)^3$$

$$s_2(x) = c(x-2)^2$$

$$s_3(x) = d(x-2)^2 + e(x-1)^3$$

We require that S, S', and S'' be continuous:

$$S\text{: continuous} \Rightarrow$$

$$a(1-2)^2 + b(1-1)^3 = c(1-2)^2 \Rightarrow a = c$$

$$c(3-2)^2 = d(3-2)^2 + e(3-3)^3 \Rightarrow c = d$$

$$S'\text{: continuous} \Rightarrow$$

$$2a(1-2) + 3b(1-1)^2 = 2c(1-2) \Rightarrow a = c$$

$$2c(x-2) = 2d(3-2)^2 + 3e(3-3)^2 \Rightarrow c = d$$

$$S''\text{: continuous} \Rightarrow$$

$$2a + 6b(1-1) = 2c \Rightarrow a = c$$

$$2c = 2d + 6e(3-3) \Rightarrow c = d$$

Thus, the only restriction we have on these values that will make $f(x)$ a cubic spline is that we require that:

$$a = c = d$$

Next, determine the values of the parameters so that the cubic spline interpolates this table:

x	0	1	4
y	26	7	25

$$f(0) = s_1(0) = a(0-2)^2 + b(0-1)^3 = 4a - b = 26$$

$$f(1) = s_2(1) = a(1-2)^2 = a = 7$$

$$\Rightarrow 4(7) - b = 26 \Rightarrow b = 2$$

$$f(4) = s_3(4) = 7(4-2)^2 + e(4-3)^3 = 28 + e = 25 \Rightarrow e = -3$$

$$\therefore\ S(x) = f(x) = \begin{cases} 7(x-2)^2 + 2(x-1)^3 & x \in (-\infty, 1] \\ 7(x-2)^2 & x \in [1,3] \\ 7(x-2)^2 - 3(x-3)^3 & x \in [3, \infty) \end{cases}$$